POLITECNICO DI TORINO

Master's Degree in Electronics

Master's Degree Thesis

Loss analysis of a 4-level,flying capacitor,bridge-less,totem-pole PFC

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Chapter 1

SPICE model Validation

In order to properly analyze more critical components in terms of losses,it is useful to first create a digital model of the device;a validation procedure of the SPICE models of each device has been performed. The following sections describe the procedure and results of this validation procedure.

All simulation results are available [1]

1.1 Fast Leg GaN FET model validation

This section describes the behavior of the SPICE model of EPC2304, compared to the EPC datasheet [2]

1.1.1 Static charateristics

Output I-V charateristics

```
.temp 25

.dc Vds 0 6 0.1 Vgs list 2 3 4 5
```

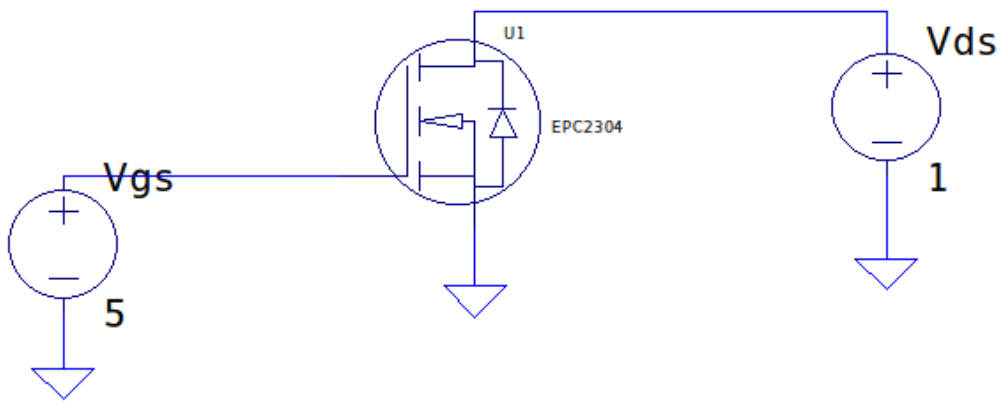

Figure 1.1: Schematic for I-V validation

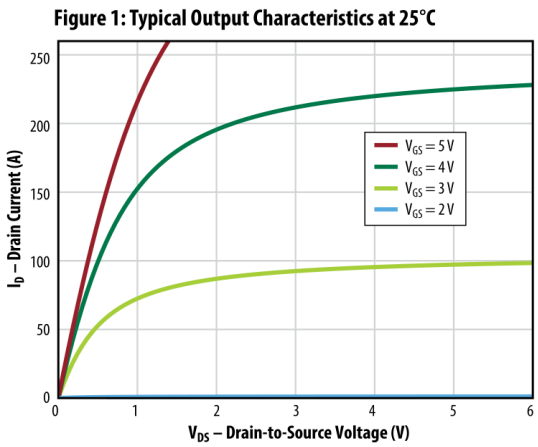

Figure 1.2: Datasheet I-V

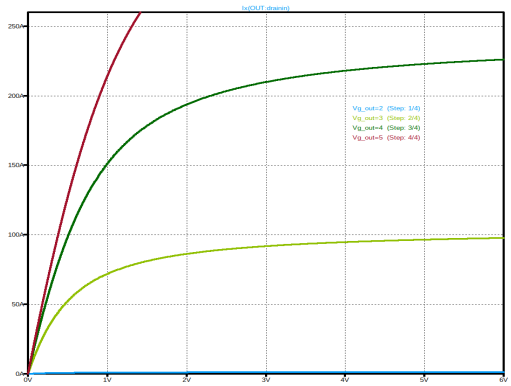

Figure 1.3: SPICE I-V

Vds@Vgs	Ids (datasheet)	Ids (simulation)
Vds=1V;Vgs=2V	$\approx 0A$	0.92A
Vds=6V;Vgs=3V	$\approx 100A$	97.69A
Vds=1V;Vgs=4V	$\approx 150A$	151.17A
Vds=1V;Vgs=5V	$\approx 215A$	213.99A

Table 1.1: I-V data comparision

Trans charateristics

```
.step temp 25 125 100

.temp 25

.dc Vgs 0.5 5 0.1
```

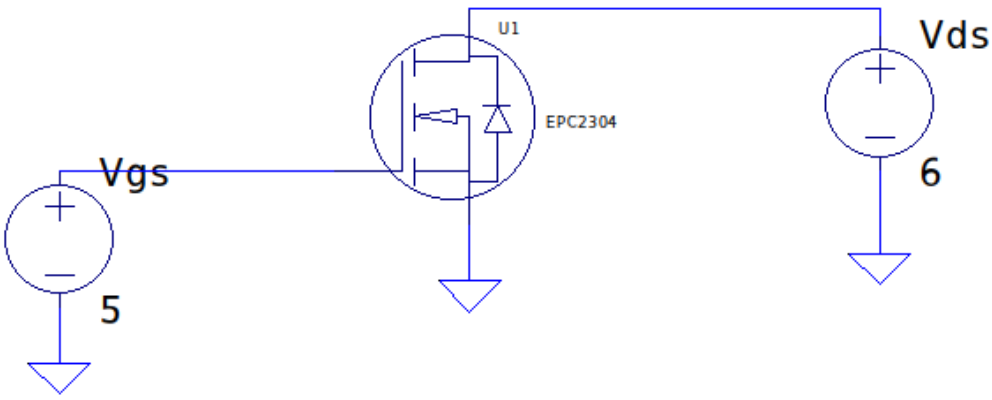

Figure 1.4: Schematic for trans charateristics validation

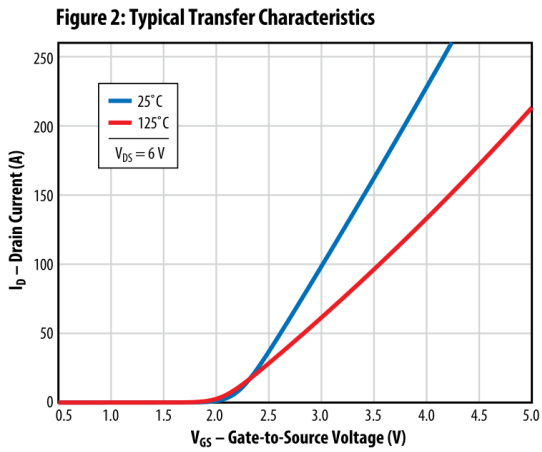

Figure 1.5: Datasheet trans charateristics

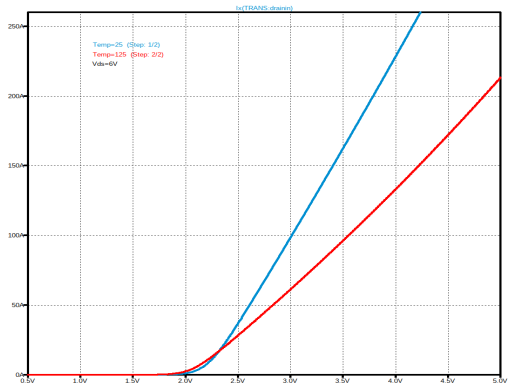

Figure 1.6: SPICE trans charateristics

Vgs@T	Ids (datasheet)	Ids (simulation)
Vgs=2.5V;T=25°C	$\approx 40A$	36.79A
Vgs=3V;T=25°C	$\approx 100A$	98.42A
Vgs=3.5;T=125°C	$\approx 100A$	96.28A
Vgs=4.5V;T=125°C	$\approx 175A$	172.04A

Table 1.2: I-V data comparision

Rdson

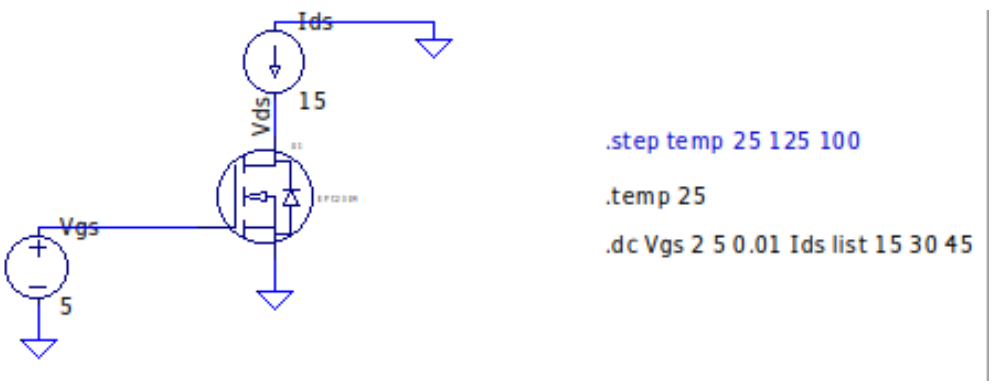

Figure 1.7: Schematic for Rdson validation

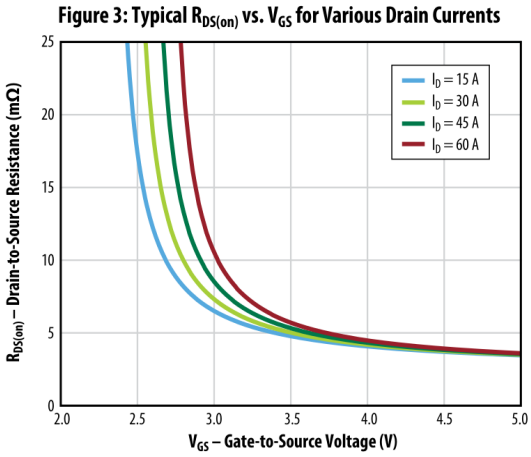

Figure 1.8: Datasheet Rdson

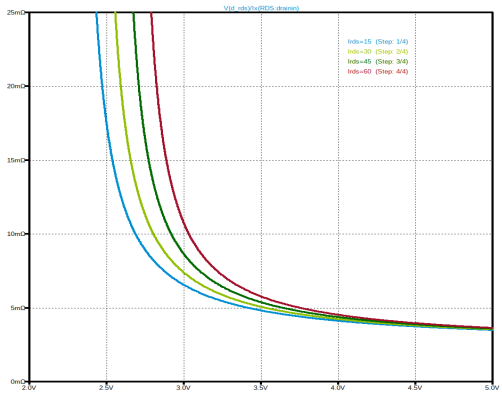

Figure 1.9: SPICE Rdson @25°C

Vgs@Ids	Rds (datasheet)	Rds (simulation)
Vgs=2.5V;Id=15A	$\approx 17.5m\Omega$	$17.46m\Omega$
Vgs=3V;Ids=60A	$\approx 10m\Omega$	$10.76m\Omega$
Vgs=3.5V;Ids=30A	$\approx 5m\Omega$	$5.073m\Omega$
Vgs=5V	$\approx 4m\Omega$	$3.57m\Omega$

Table 1.3: Rds data comparision

Rdson vs tempearture

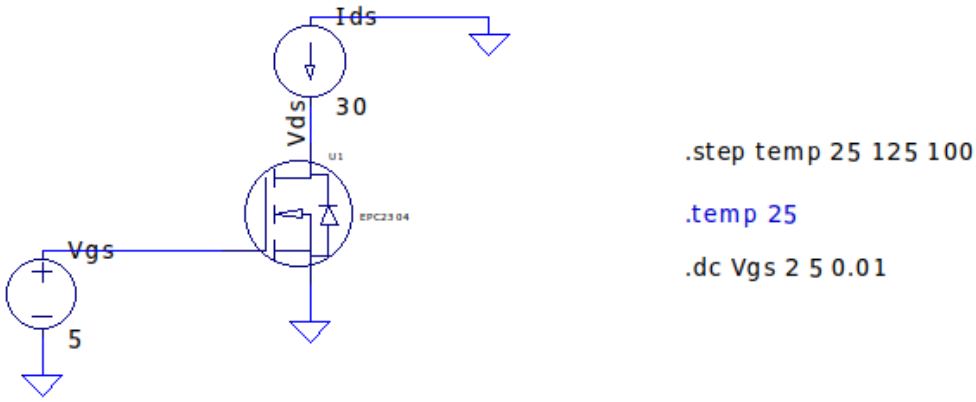

Figure 1.10: Schematic for Rdson vs tempearture

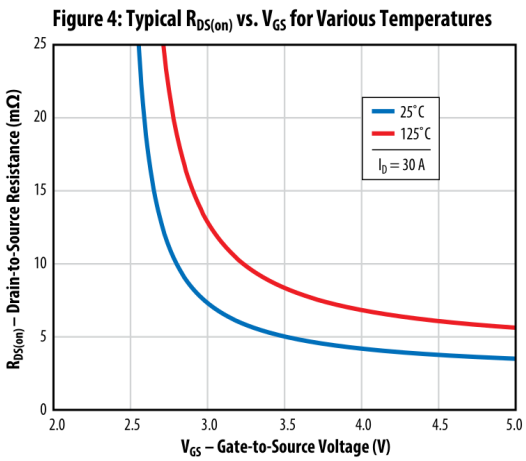

Figure 1.11: Datasheet Rdson vs tempearture

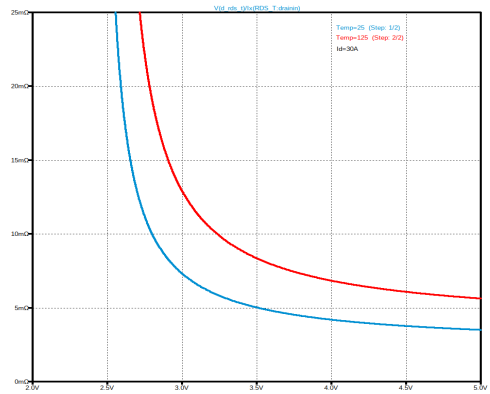

Figure 1.12: SPICE Rdson vs tempearture

Vgs@T	Rds (datasheet)	Rds (simulation)
Vgs=3.5V;T=25°C	$\approx 5m\Omega$	$5.025m\Omega$
Vgs=5V;T=125°C	$\approx 5.5m\Omega$	$5.629m\Omega$
Vgs=2.95V;T=25°C	$\approx 7m\Omega$	$7.8m\Omega$

Table 1.4: Rds temperature comparision

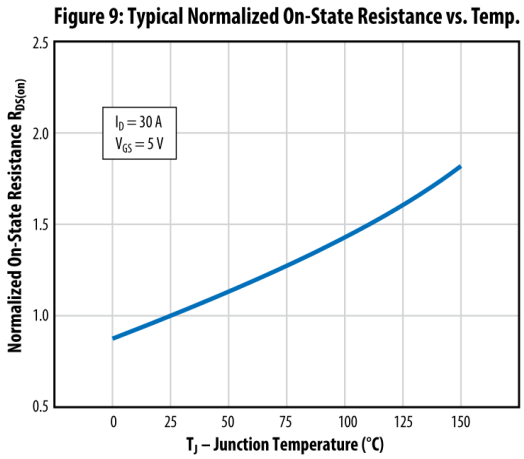

Figure 1.13: Datasheet Rdson vs temperature

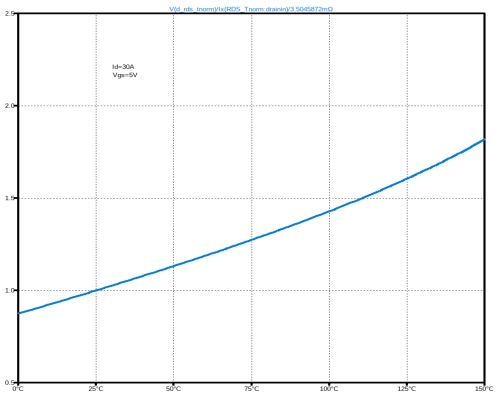

Figure 1.14: SPICE Rdson vs temperature

Reverse body diode

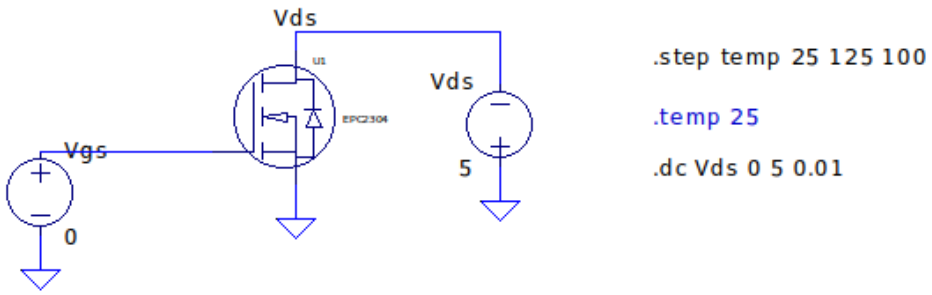

Figure 1.15: Schematic for Reverse body diode

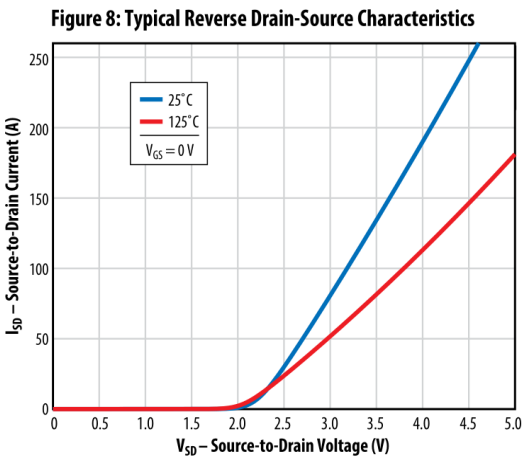

Figure 1.16: Datasheet Reverse body diode

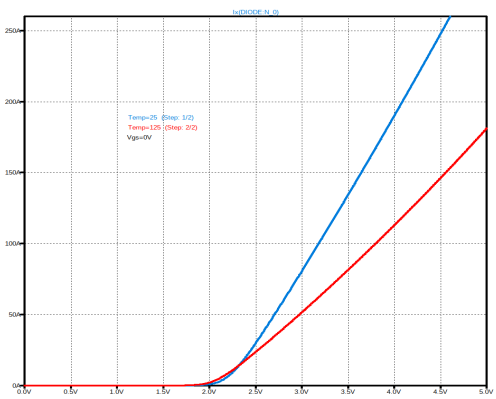

Figure 1.17: SPICE Reverse body diode

Vgs@T	Isd (datasheet)	Isd (simulation)
Vsd=4.5V;T=25°C	$\approx 250A$	$247.53A$
Vsd=3V;T=125°C	$\approx 50A$	51.71
Vsd=4.5V;T=125°C	$\approx 148A$	$146.23A$

Table 1.5: Reverse diode data comparision

Vth vs temperature

Figure 1.18: Schematic for Vth vs temperature

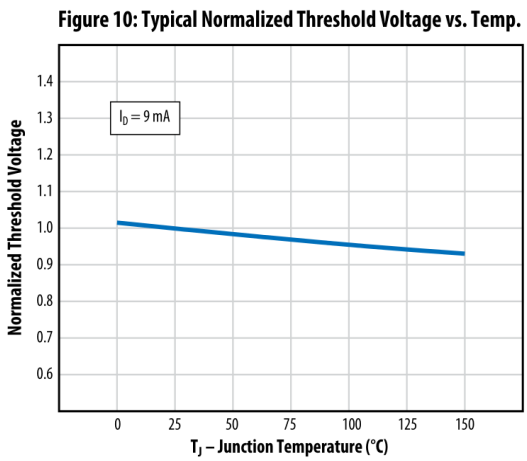

Figure 1.19: Datasheet Vth vs temperature

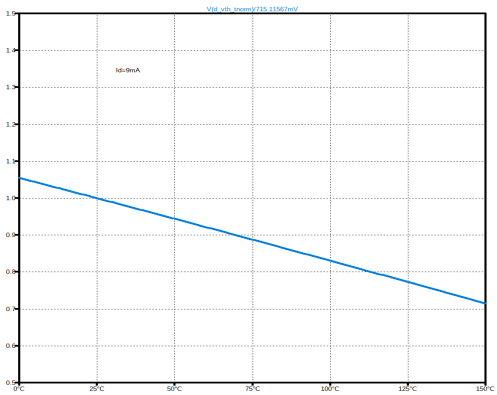

Figure 1.20: SPICE Vth vs temperature

It appears that the Vth temperature coefficents are too high compared to the datasheet one, yet for the purpose of this work is still an acceptable model.

1.1.2 Dynamic Charateristics

Typical switching curves

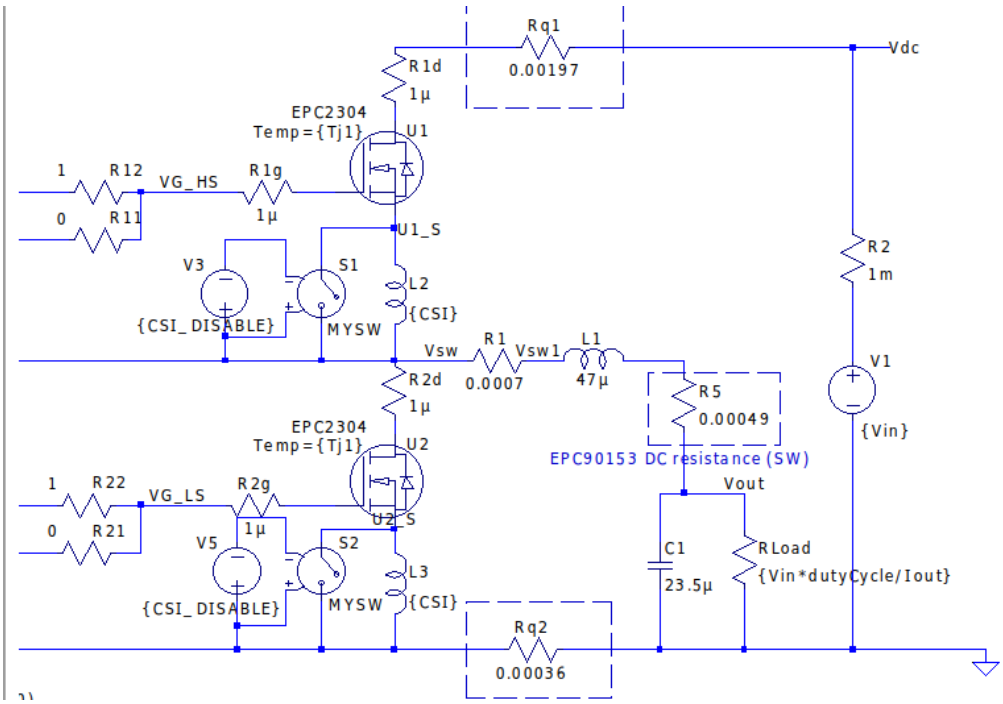

Figure 1.21: Schematic for switching behaviour

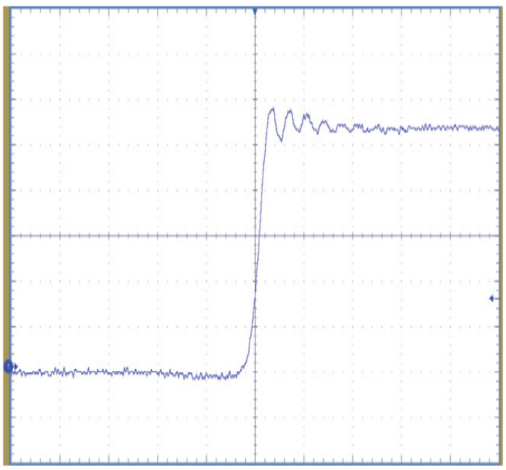

Figure 1.22: Datasheet Turn on curve

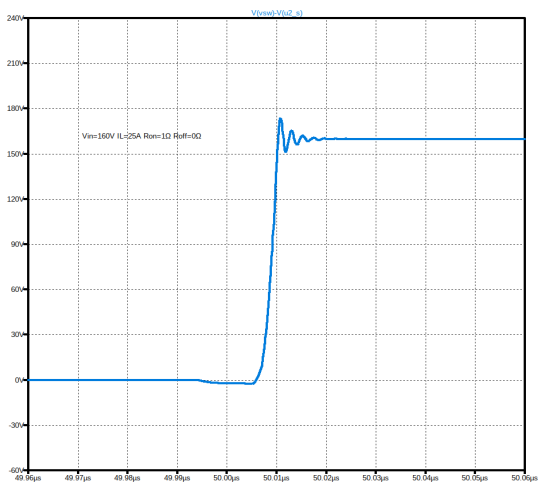

Figure 1.23: SPICE Turn on curve

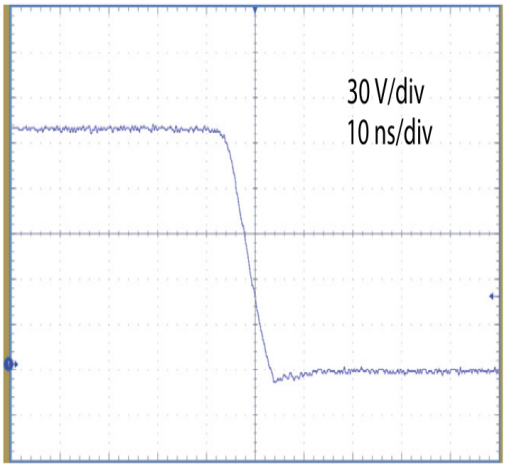

Figure 1.24: Datasheet Turn off curve

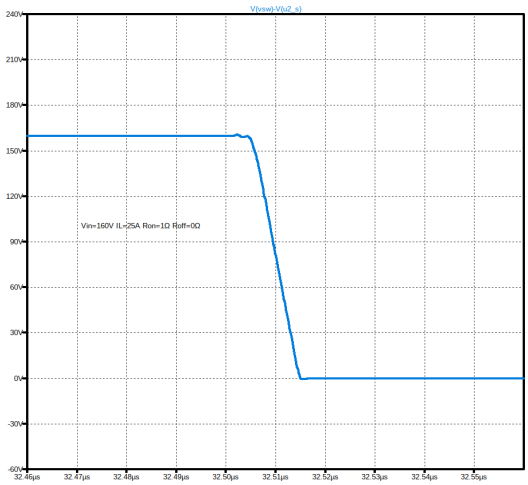

Figure 1.25: SPICE Turn off curve

Typical gate charge

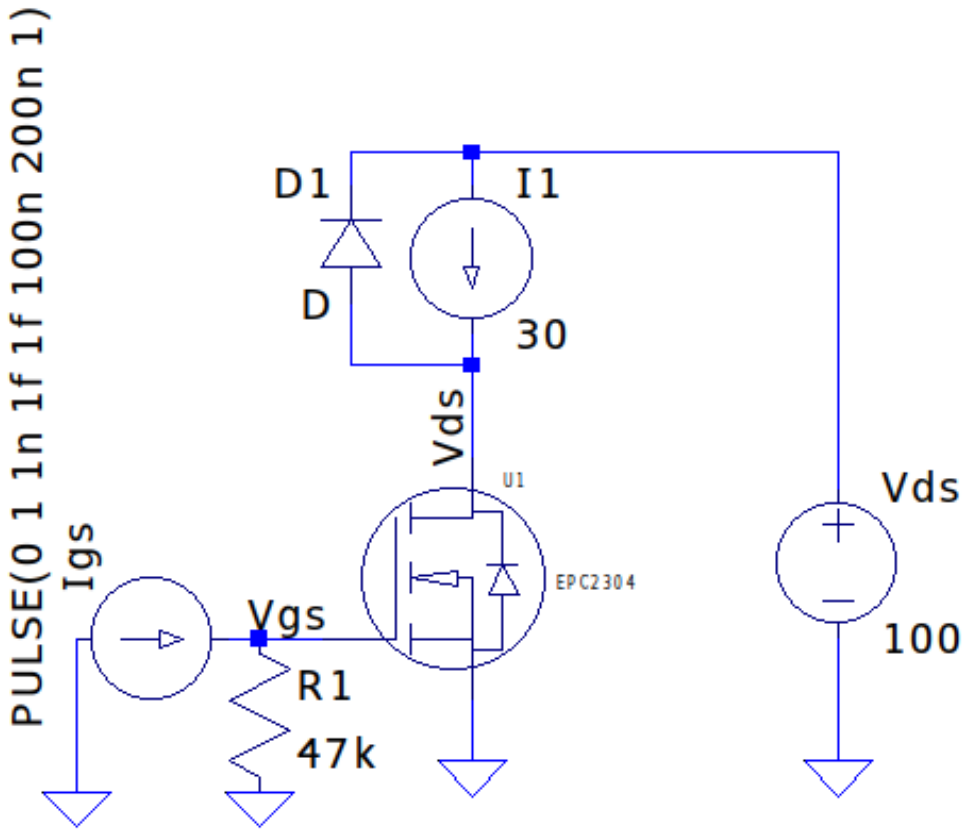

Figure 1.26: Schematic for gate charge

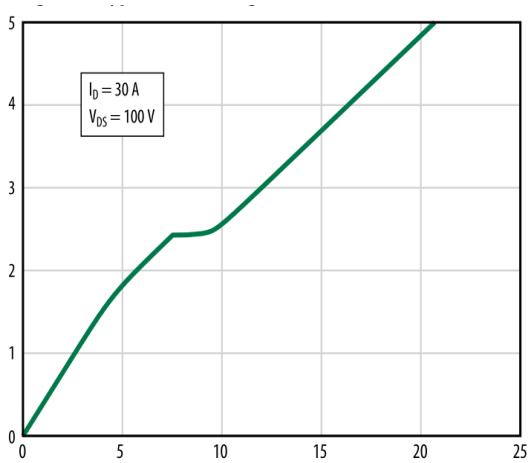

Figure 1.27: Datasheet gate charge

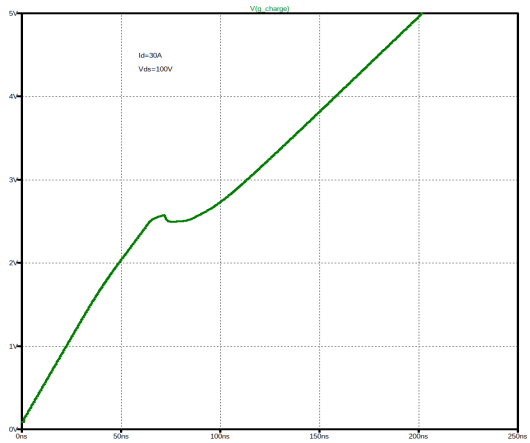

Figure 1.28: SPICE gate charge

Q (datasheet)	Q (simulation)
Qgtot=21nC	Qgtot=20.1nC
Qgs=7.5nC	Qgs=6.35nC
Qgd=2nC	Qgd=1.8nC
QVth=5.2nC	QVth=4.8nC

Table 1.6: Gate charge data comparision

Gate charges are slightly lower than datasheet one, but still in an acceptable error range,due to device variations.

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Bibliography

[1] Giorgio Armignacco. *Thesis repository*. https://github.com/arminioboy/PoliTOThesis. 2026 (cit. on p. 3).

[2] EPC-CO. *eGaN® FET DATASHEET EPC2304*. https://epc-co.com/epc/Portals/0/epc/documents/datasheets/EPC2304_datasheet.pdf. 2025 (cit. on p. 3).